Naor Larian

ELECTRICAL ENGINEERING STUDENT | FPGA / RTL INTERNSHIP

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SUMMARY

Final-year B.Sc. Electrical & Electronics Engineering student (Computers & Communications, expected 2027) seeking an **FPGA / RTL internship**. Hands-on VHDL experience across the FPGA flow, from FSM-based architecture and RTL simulation to synthesis, implementation and on-board debugging in Xilinx Vivado, demonstrated on physical Artix-7 hardware. Nearly four years of troubleshooting live satellite systems reinforce a systematic approach to debugging.

FPGA / RTL PROJECTS

- **Cyber Gladiators** – Independent FPGA Project 2026

VHDL · Basys 3 (Artix-7) · ESP32 · UART · VGA · Implemented on hardware

- Designed a **modular VHDL architecture** with **FSM-based control** for real-time 640 × 480 VGA output, sprite rendering, collision detection and game logic.
- **Integrated an ESP32 controller over UART** as the external control interface.
- **Verified RTL modules in simulation**, synthesized and implemented the design in Vivado, then **debugged the UART-to-VGA path on a physical Basys 3 board**.

- **RGB Median Filter** – Afeka Team Project 2026

VHDL · FSM · RAM/ROM · RTL Simulation · Testbenches · Simulation-verified

- Developed internal **VHDL components** for a 3 × 3 RGB median-filter architecture.
- Integrated **FSM control, data buffers and RAM/ROM handling** into the datapath.
- Verified module- and top-level behavior through **RTL simulation and testbenches** using image test data.

EXPERIENCE

- **Technical Support Engineer** Dec 2022 – Present

AMOS-Spacecom · Satellite Operator

- Troubleshoot **15–25 technical incidents per week** on live satellite communication systems, correlating RF spectrum measurements, modem indicators and link performance data to isolate faults.
- Remotely support VSAT systems and approximately **10–15 CPI** and **10–15 commissioning activities** per month, coordinating with customers and engineering teams.
- Develop and refine **technical SOPs** that improve troubleshooting consistency and make critical procedures easier to find.

EDUCATION

Expected 2027
B.Sc. Electrical & Electronics Engineering
Afeka Tel Aviv Academic College of Engineering
Specializations: Computers & Communications

2022
Practical Engineer Diploma, Electrical Engineering
Handessaim Tel Aviv
Graduated with honors

TECHNICAL SKILLS

HDL & RTL Design
VHDL, FSM design, modular RTL architecture, datapath integration

Verification
RTL simulation, testbenches

FPGA Flow & Debug
Xilinx Vivado (synthesis, implementation), on-board debugging

Hardware & Interfaces
Artix-7 / Basys 3, ESP32, UART, VGA, RAM/ROM

Programming & Tools
C, Python, MATLAB (academic); spectrum analyzer (professional)

RELEVANT COURSEWORK

Advanced Hardware Design, Digital Signal Processing, Signals & Systems, Communication Systems

LANGUAGES

Hebrew – Native
English – Professional working proficiency